Light

GCSE Computer Science — Programming Data Structures (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a variable in programming?

- ○ **A.** A fixed value that cannot be changed during program execution
- ○ **B.** A named location in memory that stores a value which can change while the program runs
- ○ **C.** A type of loop that repeats a set number of times
- ○ **D.** A section of code that performs a specific task

Q2 (1 mark)

Why would a programmer use a constant instead of a variable?

- ○ **A.** To store a value that should not change during program execution, such as VAT_RATE = 0.20
- ○ **B.** To store user input that will be processed later
- ○ **C.** To count how many times a loop has executed
- ○ **D.** To store the result of a calculation that will be displayed

Q3 (1 mark)

Which data type would be most appropriate to store the number of students in a class?

- ○ **A.** Real
- ○ **B.** String
- ○ **C.** Integer
- ○ **D.** Boolean

Q4 (1 mark)

What values can a Boolean variable store?

- ○ **A.** Any whole number
- ○ **B.** Any text characters
- ○ **C.** Numbers with decimal places
- ○ **D.** Only True or False

Q5 (1 mark)

A program needs to store the price of an item as £12.99. Which data type should be used?

- ○ **A.** Integer
- ○ **B.** Real
- ○ **C.** Boolean
- ○ **D.** Character

Q6 (1 mark)

What is an array?

- ○ **A.** A single variable that can only hold one value
- ○ **B.** A type of loop used to repeat code
- ○ **C.** A function that returns multiple values
- ○ **D.** A data structure that stores multiple values of the same data type under one identifier

Q7 (1 mark)

An array is declared as: `names = ['Ali', 'Ben', 'Cat', 'Dan']`
What value is stored at index 0?

- ○ **A.** Ben
- ○ **B.** Cat

- ○ **C.** Dan
- ○ **D.** Ali

Q8 (1 mark)

Given `word = 'Hello'`, what is the output of `word.length`?

- ○ **A.** 4
- ○ **B.** 6
- ○ **C.** 5
- ○ **D.** Hello

Q9 (1 mark)

Given `text = 'computer'`, what is the output of `text.upper`?

- ○ **A.** COMPUTER
- ○ **B.** computer
- ○ **C.** Computer
- ○ **D.** 8

Q10 (1 mark)

What does casting mean in programming?

- ○ **A.** Deleting a variable from memory
- ○ **B.** Converting a value from one data type to another
- ○ **C.** Creating a copy of an array
- ○ **D.** Combining two strings together

Q11 (1 mark)

What is the result of concatenating 'Good' and 'Morning'?

- ○ **A.** GoodMorning
- ○ **B.** Good Morning
- ○ **C.** Morning Good
- ○ **D.** Good, Morning

Q12 (1 mark)

Which data type is used to store a single letter or symbol?

- ○ **A.** String
- ○ **B.** Integer
- ○ **C.** Character
- ○ **D.** Boolean

Q13 (2 marks)

A program stores test scores in an array: `scores = [85, 92, 78, 90, 88]`
Which code correctly calculates the total of all scores?

- ○ **A.**
```
total = 0
for i = 0 to 4
  total = total + scores[i]
next i
```
- ○ **B.**
```
total = 0
for i = 1 to 5
  total = total + scores[i]
next i
```
- ○ **C.** `total = scores[0] + scores[5]`
- ○ **D.** `total = scores.length`

Q14 (2 marks)

Given `code = 'AB12345'`, what is the output of `code.left(2) + code.right(3)`?

- ○ **A.** AB123
- ○ **B.** AB345
- ○ **C.** 12AB
- ○ **D.** 45AB

Q15 (2 marks)

A user enters '25' when prompted for their age. The program needs to check if they are over 18.
Why is casting necessary before performing this check?

- ○ **A.** To make the program run faster
- ○ **B.** To display the age on screen
- ○ **C.** To store the age in an array
- ○ **D.** Because input is received as a string and must be converted to an integer before numeric comparison

Q16 (2 marks)

A 2D array is declared as:
`grid = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`
What value is returned by `grid[1][2]`?

- ○ **A.** 2
- ○ **B.** 4
- ○ **C.** 6
- ○ **D.** 8

Q17 (2 marks)

Given `email = 'user@school.org'`, which operation extracts 'school'?

- ○ **A.** `email.substring(5, 6)`
- ○ **B.** `email.left(6)`
- ○ **C.** `email.right(6)`
- ○ **D.** `email.substring(5, 10)`

Q18 (2 marks)

An array contains an unknown number of elements. Which loop correctly processes every element?

- A.
```
for i = 0 to 10
  print(data[i])
next i
```
- B.
```
for i = 0 to data.length - 1
  print(data[i])
next i
```
- C.
```
for i = 1 to data.length
  print(data[i])
next i
```
- D.
```
for i = 0 to data.length + 1
  print(data[i])
next i
```

Q19 (2 marks)

What is the output of the following code?

```
num = '10'
result = int(num) + 5
print(result)
```

- A. '105'
- B. '10' + 5
- C. 15
- D. Error

Q20 (2 marks)

A login system needs to accept usernames regardless of how the user types them (e.g., 'ADMIN', 'admin', 'Admin' should all match).
Which approach correctly implements case-insensitive comparison?

- ○ **A.** Compare the strings directly without any conversion
- ○ **B.** Use .length to check if the usernames have the same number of characters
- ○ **C.** Convert the username to a number before comparing
- ○ **D.** Convert both the input and stored username to the same case (e.g., .lower) before comparing

Q21 (2 marks)

A school needs to store each student's name, age, and form group together as a single unit.
Which data structure is most appropriate?

- ○ **A.** A record with fields for name, age, and form group
- ○ **B.** Three separate unrelated variables
- ○ **C.** A single string containing all the information
- ○ **D.** Three separate arrays, one for each piece of data

Q22 (2 marks)

Given the array `prices = [10, 20, 30, 40, 50]`, which statement correctly changes the third element to 35?

- ○ **A.** prices[3] = 35
- ○ **B.** prices[2] = 35

- ○ **C.** prices(3) = 35
- ○ **D.** prices.set(3, 35)

Q23 (2 marks)

A program needs to display: 'Your score is 85 points'
Given `score = 85` (an integer), which code produces the correct output?

- ○ **A.** print('Your score is ' + score + ' points')
- ○ **B.** print('Your score is 85 points')
- ○ **C.** print('Your score is ' + str(score) + ' points')
- ○ **D.** print('Your score is ', score, ' points')

Q24 (3 marks)

A program searches for a name in an array of 1000 student names.
`students = ['Ali', 'Ben', ... , 'Zoe'] // 1000 names`

```
searchName = 'Emma'
found = False
for i = 0 to 999
  if students[i] == searchName then
    found = True
  endif
next i
```

Analyse the efficiency problem with this algorithm and explain how it could be improved.

- ○ **A.** The algorithm is already optimal and cannot be improved

○ **B.** The algorithm should use a 2D array instead of a 1D array

○ **C.** The algorithm should start searching from the end of the array instead

○ **D.** The algorithm continues searching all 1000 elements even after finding the name; adding a check to exit the loop once found would improve efficiency by avoiding unnecessary comparisons

Q25 (3 marks)

A 2D array stores a 3×4 grid of temperature readings:

```
temps = [[20, 22, 21, 23],
         [19, 21, 20, 22],
         [18, 20, 19, 21]]
```

Analyse what nested loops are required to calculate the total of all temperature readings.

○ **A.** A single loop from 0 to 11 accessing temps[i]

○ **B.** A nested loop structure where the outer loop iterates through rows (0 to 2) and the inner loop iterates through columns (0 to 3), accessing temps[row][col] and adding each value to a running total

○ **C.** Two separate sequential loops, one for rows and one for columns

○ **D.** A loop that only processes the diagonal elements

Q26 (3 marks)

A password validation algorithm must check that a password:

- Is at least 8 characters long
- Contains at least one digit (0-9)

Analyse which combination of string operations would implement these checks.

- ○ **A.** Only check password.length >= 8
- ○ **B.** Use password.length >= 8 for length check, then iterate through each character using a loop with password.substring(i, 1) or indexing, checking if each character is between '0' and '9' to find a digit
- ○ **C.** Use password.upper to check for digits
- ○ **D.** Use password.right(1) to check if the password ends with a digit

Q27 (3 marks)

A library system stores book information using records:

```
book record:
  title: string
  author: string
  available: boolean
  copies: integer
```

The system stores 500 books in an array of records called `library`. Analyse how to find all available books by a specific author.

- ○ **A.** Check only library[0].author and library[0].available
- ○ **B.** Use library.author to get all authors at once
- ○ **C.** Iterate through the array from index 0 to 499, and for each book check if library[i].author matches the search author AND library[i].available is True; when both conditions are met, the book should be included in results
- ○ **D.** Sort the array by author name first, then search

Q28 (3 marks)

A game developer needs to store a player's health value that ranges from 0 to 100.
Analyse the trade-offs between using an integer versus a real (float) data type for this purpose.

- ○ **A.** Real is always better because it can store more precise values
- ○ **B.** There is no difference between integer and real for this use case
- ○ **C.** Integer is preferable because health is a whole number in this range, integers use less memory, comparisons are faster and exact, and there are no rounding errors; real would only be needed if fractional health values (like 75.5) were required
- ○ **D.** Real is required because 100 is too large for an integer

Q29 (3 marks)

A program receives date input in the format 'DD/MM/YYYY' (e.g., '25/12/2024').
Analyse how string operations could extract the day, month, and year as separate integer values.

- ○ **A.** Use the date string directly in calculations without extraction
- ○ **B.** Use .left(2) to get day as string, .substring(3, 2) to get month as string, .right(4) to get year as string, then cast each to integer using int() for numerical operations
- ○ **C.** Use .upper to convert the date to numbers
- ○ **D.** Use .length to determine which part is the day, month, or year

Q30 (3 marks)

A quiz program needs to store 20 questions and their corresponding correct answers.
Two design approaches are proposed:
Design A: Two parallel arrays - `questions[20]` and `answers[20]`
Design B: An array of 20 records, each containing question and answer fields
Analyse the advantages and disadvantages of each approach.

- ○ **A.** Design A is always better because arrays are simpler than records
- ○ **B.** Design B is always better because it uses less memory
- ○ **C.** Design A (parallel arrays) is simpler to implement but risks index synchronisation errors if arrays are modified; Design B (array of records) keeps related data together, making code more maintainable and less error-prone when adding, deleting, or reordering questions, though it requires understanding of record structures
- ○ **D.** Both designs are identical in every way

Total: **55** marks • Time: **30 min**

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